
A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We turn a language model into a *cellular automaton over token space* — a ring
2 of token cells resampled in place by the model’s own windowed conditional
3 $p_r(x_i \mid x_{i\pm r})$, probed by common-random-number (CRN) damage spreading —
4 and develop it into a *validated* measurement instrument. The organizing princi-
5 ple is **validation by reproduction**. On the Domany–Kinzel stochastic automaton,
6 where the damage field is provably the automaton itself, an independent predic-
7 tion matches ours **bit-exactly** — zero mismatching cells over 4096×1500 up-
8 dates, with a nonzero off-line control — calibrating the apparatus with no error
9 bar attached; the instrument also separates elementary CA rules by whether dam-
10 age survives ($d=3.0$) and recovers known Markov transition matrices. Applied
11 across Pythia-410m training checkpoints it detects a **developmental transition**:
12 the token-space Lyapunov exponent goes from seeds disagreeing about its *sign* to
13 **every one of 48 post-transition runs being positive**, at two lattice sizes, with all
14 four members of a pre-registered family surviving Benjamini–Hochberg correc-
15 tion. We then *delimit* the instrument, which is the result we consider most useful:
16 real autoregressive generation absorbs *no* injected token error ($P_{\text{persist}}=1.000$ on
17 three models), because free generation never resamples a token. These quanti-
18 ties characterise an iterated-resampling construction, not deployed decoding — a
19 boundary we draw rather than leave to a reader.

20 1 Introduction

21 Language models are evaluated almost entirely as functions: given a context, how good is the next-
22 token distribution? But a masked or autoregressive conditional also *defines a dynamics* — iterate it
23 in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the
24 model. Iterated-map views of LLMs have appeared qualitatively [Utimula, 2026, Perez et al., 2024,
25 Wang et al., 2025] and as mixing-time studies of masked-LM Glauber dynamics [Sana et al., 2026];
26 the phenomena a CA vocabulary names — damage spreading, light cones, Lyapunov exponents,
27 attractor censuses [Hanson and Crutchfield, 1997] — are decades old. What has been missing is not
28 a phenomenon but a *measurement instrument whose readings can be trusted*.

29 Our organizing principle is therefore **validation by reproduction** (§3): before turning the apparatus
30 on a language model we require it to recover values that are independently known. This is not a for-
31 mality — building the ladder forced us to retract several of our own earlier claims, and the strongest
32 rung is one where agreement is *exact* rather than fitted, so it cannot be passed by a plausible-looking
33 estimator. We use only *simple, mature* primitives [Bagnoli et al., 1992, Lieb and Robinson, 1972,

34 [Hanson and Crutchfield, 1997](#)] rather than bespoke machinery, precisely because their behaviour
 35 on known systems is a fixed benchmark. Only then do we report what the instrument measures on
 36 trained LMs (§4) and, equally, where its readings provably stop applying (§5, §6). The criticality
 37 ideas the results touch are old [[Langton, 1990](#), [Bertschinger and Natschläger, 2004](#), [Zhang et al., 2025](#)];
 38 our contribution is a calibrated black-box *measurement* of them and an explicit statement of
 39 its limits.

40 2 The instrument

41 **Rule.** A window of $w=2r+1$ tokens with the centre masked is fed to a masked LM; the tempered
 42 centre distribution is the kernel $p_r(x_i | x_{i\pm r})$. For an autoregressive model the window is the r cells
 43 to the left. **Automaton.** N ring cells, async random-order single-site Glauber; special tokens forbid-
 44 den as emissions. **CRN coupling.** Twin lattices sharing initial state, update order and the uniform
 45 stream differ only in the factor under test, so the *null* arm must diverge by exactly zero — asserted on
 46 every backend in the test suite, because every damage number depends on it. **Confounds.** A lattice
 47 looping one corpus n -gram scores overlap for free, so structure uses a repetition-robust variant; a
 48 degenerate low-entropy lattice snaps a perturbation back trivially, so damage D is normalised by the
 49 model’s own diversity floor D_0 (twins under *independent* noise, no flip), giving $D_{\text{norm}}=D/D_0$. A
 50 free control disciplines our language: the temperature “phase transition” is, on a finite-size scan, a
 51 *crossover* ($\chi_{\text{peak}} \propto 1/N$). Two further readings, reported here because §5 bounds what they mean:
 52 an **effective interaction radius**, where repetition-robust long-range corpus consistency peaks at
 53 $r \approx 4$ and falls by $r=16$; and a **damage light cone** whose front velocity grows with r and is *model-*
 54 *invariant* — the classic CA light cone [[Bagnoli et al., 1992](#), [Lieb and Robinson, 1972](#), [Liu et al., 2021](#)],
 55 imported, not claimed.

56 3 Validation: reproducing known metrics

57 We check the apparatus against systems where the answer is known, and are explicit about which
 58 properties each rung does and does not exercise (Fig. 1). The ladder is ordered by how much of the
 59 instrument’s regime the rung shares: smooth-limit arithmetic first, then *discrete* state with a finite
 60 perturbation—the regime the token instrument occupies.

61 **(1–2) Smooth-limit unit tests, labelled as such.** A renormalized tangent estimator recovers the
 62 logistic map’s analytic exponent to machine precision, and a coupled-map lattice matches an exact
 63 Benettin/Jacobian reference to 1.1×10^{-3} . Neither is evidence about damage spreading: the twin
 64 is re-anchored to the reference orbit each step, so the estimator *is* a finite-difference evaluation of
 65 the derivative it is compared against, and a **token flip is $O(1)$ — there is no $\epsilon \rightarrow 0$ limit in token**
 66 **space.**

67 **(3) Elementary CA: a coarse split, decisively.** Driving the identical protocol with 19 rules of
 68 *known* class (12 seeds, rule as the unit of analysis) separates rules whose damage *survives* from
 69 those whose damage *dies*. The discriminating quantity is the **ignition probability**, not λ : damage in
 70 a discrete CA is bimodal, so a λ averaged over ignited and extinguished runs measures the mixture.
 71 It is also the directed-percolation order parameter [[Grassberger, 1995](#)]. Ordered rules sit at 0.046
 72 (CI [0.000, 0.102]) against 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest** $p=0.0$, $d=3.03$. The
 73 three-class ordering does *not* survive ($p=0.470$), so we claim only the coarse split — and we report
 74 this rather than the group-mean λ ordering we first computed, whose ordered end is not a measure-
 75 ment: five of seven ordered rules return exactly $-0.4 \ln 10$, the constant a log-linear fit yields when
 76 damage dies immediately.

77 **(4) Domany–Kinzel: an exact check on the damage machinery.** The DK automaton [[Domany](#)
 78 [and Kinzel, 1984](#)] is the only rung that is stochastic *and* discrete—the instrument’s own regime.
 79 On its $p_2=0$ line the damage field between two commonly-driven replicas is provably *itself* a DK
 80 automaton at the same p_1 [[Kohring and Schreckenberg, 1992](#)]: with $s'_i = (s_{i-1} \oplus s_{i+1}) \theta(p_1 - z_i)$,
 81 sharing z_i gives $d'_i = (d_{i-1} \oplus d_{i+1}) \theta(p_1 - z_i)$. We therefore predict the damage trajectory with an
 82 independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \dots, 1.0\}$ on a ring of 4096 over
 83 1500 steps, **zero mismatching cells**, with 16 in an off-line control at (0.6, 0.5) where the mapping
 84 must break. This runs through the same simulation loop that produces every language-model number
 85 below, so the window indexing, the shared-uniform consumption order, the sampling and the update

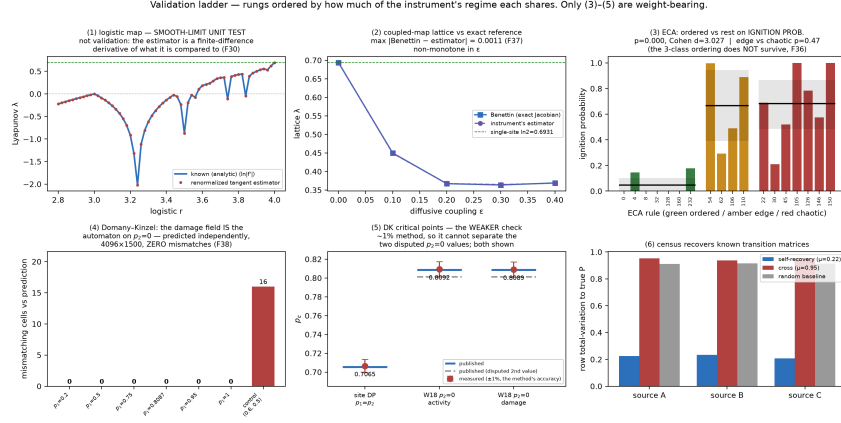


Figure 1: Validation ladder, ordered by how much of the instrument’s regime each rung shares; only (3)–(5) are weight-bearing. Panel (4) is the load-bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, with the off-line control beside it showing the test is not vacuous.

86 schedule are all verified with no error bar attached. A second closed form rides inside the same run:
 87 at $p_1=1$ the update is deterministic $\oplus$ (Rule 90), so the damage field must hold exactly $2^{\text{popcount}(t)}$
 88 live cells—at $t=1500$, $128/4096 = 0.03125$, which is what we obtain. Critical points are a weaker,
 89 $\sim 1\%$ check: $p_c=0.7065$ on the site-DP line (published $0.705489(4)$) and 0.8092 on $p_2=0$, with
 90 the damage field measured *independently* at 0.8089 — coinciding with the activity transition to
 91 3×10^{-4} , as the mapping requires. The published $p_2=0$ value is itself disputed ($0.801(2)$ [Zebende
 92 and Penna, 1994] vs. $0.8087(5)$ [Hinrichsen et al., 1997]) and our accuracy cannot separate them;
 93 we report both and claim neither.

94 **(5) Synthetic Markov sources.** The attractor census, run on tiny models trained on three sources
 95 with known sparse matrices, recovers each model’s own matrix (row total-variation 0.22) and not
 96 the others (cross 0.95 ; random baseline 0.91). Rungs (3)–(5) share the instrument’s regime and are
 97 what license its use on language models; (4) is the only one on which agreement is exact rather than
 98 fitted.

99 4 A developmental transition in token-space criticality

100 Applied across Pythia-410m checkpoints (step256–step143000, 8 seeds, lattice sizes $N=48$ and
 101 96; Fig. 2), the calibrated instrument finds the model’s token-space dynamics changing from sub-
 102 critical to super-critical between steps 512 and 2000, and staying there for the rest of training.

103 We pre-registered the test (post- vs pre-transition checkpoints, run-level Mann–Whitney, family
 104 $= \{\lambda_{ca}, D_{\text{norm}}\} \times \{N=48, 96\}$, BH-FDR and a demotion rule: failure at either size demotes the
 105 headline. **All four members survive** ($p_{\text{BH}} \leq 2 \times 10^{-5}$). From the pre-registered pre set (256/512)
 106 to the settled plateau (2000/8000/143000), λ_{ca} moves $+0.0247 \rightarrow +0.1683$ at $N=48$ ($d=1.59$)
 107 and $+0.0320 \rightarrow +0.1686$ at $N=96$ ($d=1.71$, $n_{\text{pre}}=15$: one run is excluded because its damage
 108 never ignited, so λ_{ca} is undefined for it).

109 **The claim is better stated without any pre/post split**, since every effect size depends on where
 110 that line is drawn. Before the transition the seeds do not agree on the *sign* of λ_{ca} (6/16 negative
 111 at $N=48$, 7/15 at $N=96$, spanning -0.216 to $+0.320$); after it, **not one of 48 plateau runs is**
 112 **negative**, the minimum being $+0.107$. That uses every run, is ordinal, and survived a later change
 113 to how unignited runs are handled that moved every mean in the analysis.

114 **The curve is not a step, and both ends of the contrast can be inflated.** It overshoots at step
 115 1000 by $1.4\text{--}22.4\%$ and settles; taking that peak as the level, or the lowest checkpoint alone as the
 116 baseline, inflates the effect ($1.7\times$ on λ_{ca} for the latter). We use the pre-registered sets at both ends.
 117 The overshoot survives correction in one of four cells, and that cell is D_{norm} — on λ_{ca} it is $+1.4\%$

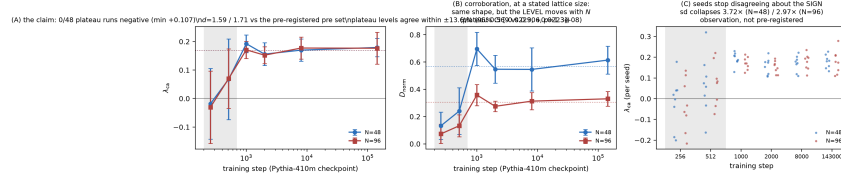


Figure 2: The developmental transition. (A) λ_{ca} rises and settles positive, the two lattice sizes on top of each other (levels agree within $\pm 14\%$). (B) D_{norm} : same shape, size-dependent level — why it corroborates rather than carries. (C) Per-seed values: sign disagreement before, consensus after.

118 at $N=96$ ($p_{BH}=0.78$) — so non-monotonicity is largely a D_{norm} phenomenon and step 10000 is not
 119 claimed as a distinct phase.

120 **Which quantity carries the claim.** λ_{ca} is size-robust and D_{norm} is not, and a third lattice size settles
 121 why. Over $N=48/96/192$ the λ_{ca} plateau is 0.168/0.169/0.160 — a log-log slope of $N^{-0.04}$, i.e.
 122 **intensive across a $4\times$ range**, varying by 5% of its mean — while D_{norm} is 0.569/0.306/0.139, a
 123 slope of $N^{-1.02}$, i.e. $1/N$. That is what the construction predicts: λ_{ca} is a cone-growth *rate* fitted
 124 before saturation, so it is intensive; D_{norm} is a density ratio whose numerator stays localised in a
 125 cone while its denominator is delocalised over every site. (At two sizes we could only state a bound:
 126 the plateau levels differed by -0.0003 , 95% CI $[-0.023, +0.022]$. We give the bound rather than a
 127 non-significant p -value, since a null result is not evidence of equality.) So λ_{ca} carries the claim and
 128 D_{norm} corroborates at a stated lattice size, never as a lattice-free property. Seed-to-seed variance
 129 also collapses $\sim 3\times$ across the transition (Levene $p \leq 1.7 \times 10^{-4}$ at both sizes); not pre-registered,
 130 reported as an observation, and note it is the checkpoint means and the two lattice sizes that agree to
 131 a few percent — the plateau per-seed CV is 22–25%.

132 5 What the damping length measures: free generation absorbs nothing

133 The instrument’s central assumption — that a number measured on the ring says something about
 134 the model — is one we tested rather than assumed, by running the identical damage protocol on *real*
 135 *autoregressive generation*: two CRN twins from a prompt sharing one uniform stream, one forced
 136 to a different token at one position, both then continuing freely. The null arm is asserted to diverge
 137 by exactly zero, and does.

138 The result is unambiguous and negative. On pythia-70m, -160m and -410m (32 trials each) the
 139 injected error **persists in every single trial** ($P_{persist}=1.000$). Token identity is harsh once an injec-
 140 tion shifts alignment, so we also measured distributionally: the total-variation distance between the
 141 twins’ next-token distributions, normalised by the TV between two *independent* continuations of the
 142 same prompt, settles at $TV_{norm} \approx 0.97$ — the twins end as far apart as unrelated continuations.

143 The mechanism is structural. **Free generation never resamples a token:** once emitted, an error
 144 stays in context permanently. The ring CA revisits every site, and in-place resampling is what creates
 145 the possibility of healing. The damping length, D_{norm} and the light cone therefore characterise *the*
 146 *iterated-resampling construction*, and we do not offer them as predictions about decoding. This
 147 also explains a light-cone velocity that depends on r but not the model, the model-invariance of
 148 $\lambda_{ca}(r)$, and the failure in §6. It does *not* touch §4, and the reason is worth stating: the construction
 149 is held *fixed* across checkpoints — same ring, radius, temperature, coupling, seeds — so a change
 150 measured across checkpoints cannot come from the apparatus. A fixed instrument reading differently
 151 on different models is what an instrument is for; what this result forbids is transporting the units to
 152 a process the instrument does not implement.

153 **Coupling, stated precisely.** Damage spreading is a property of (*model, coupling*), not of the model
 154 alone — a published result in stochastic PCAs [Kohring and Schreckenberg, 1992, Grassberger,
 155 1995]. Our inverse-CDF sampling from a shared uniform is the *monotone* coupling: on DK’s binary
 156 alphabet it coincides with the maximal-correlation member of the admissible family [Hinrichsen
 157 et al., 1997], which is why the identity above is exact, but on a vocabulary it does not, so our LM
 158 damage numbers sit inside the family rather than at an extremum (measured excess disagreement
 159 1.3–5.4%; see the supplement). What justifies the choice is replica-independence — each replica’s

next state depends on its own state and the shared noise alone — which a pairwise maximal coupling lacks. Relative comparisons, including §4, are unaffected: the coupling is a common mode.

6 The instrument’s boundary: token- vs. activation-space

A black-box instrument earns its keep if its readings track something one would otherwise need internals to see. Tested across two model families and two within-model axes, they do not: pairing λ_{ca} against a finite-depth top-Lyapunov exponent of the residual stream is family-dependent (Pythia $r=+0.71$, GPT-2 $r=-0.43$; the pooled $p=0.025$ is a Simpson’s-paradox artifact), the temperature axis is confounded by a common cause, and the radius axis is null because $\lambda_{ca}(r)$ is model-invariant and so carries no model signal to proxy. The reason is structural and compounds a regime mismatch: the white-box exponent is *architectural* (flat across training, $\approx 1/L$) and is a *tangent* quantity, while λ_{ca} is *finite* — and our own ϵ -sweep shows those two disagree even where ground truth is known. We report this as a clean **negative**: token-space and activation-space criticality are distinct, non-proxying levels.

7 Related work

Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our novelty is the measurement layer, not the framing. Edge-of-chaos measurement is canonical [Langton, 1990, Bertschinger and Natschläger, 2004] and already applied to transformers [Tomihari and Karakida, 2025] and LLMs [Zhang et al., 2025, Li et al., 2025]; we claim neither the concept nor the criticality $\leftrightarrow$ capability link. Temperature-driven order–disorder transitions in LLM generation are already studied in several independent works [Ruan et al., 2026, Sun and Haghighat, 2025, George et al., 2025, Nakaishi et al., 2024, Arnold et al., 2024]; we claim only a calibrated route to the quantity, not the quantity. Perturbation propagation in *activation* space [Petrova and Vejsiu, 2026, Olivieri and Pérez Rodríguez, 2026, Fernando and Guitchounts, 2026] and over compound-AI graphs [Nilayam and Nayak, 2026] is adjacent, in a different space. We avoid “repair” [McGrath et al., 2023, Rushing and Nanda, 2024] and “self-correction” [Petrova and Vejsiu, 2026], both categorically different from a spatial damping length. Tang et al. [2026] is the calibration kin; the census descends from Hanson and Crutchfield [1997].

8 Limitations

Neither construction samples the model’s actual distribution — masked conditionals are globally inconsistent, and the autoregressive rule, though consistent, is resampled in place on a ring — so all claims are properties of the dynamics. Rings are small ($N \leq 384$); the developmental result rests on one model family, and a second family — not a third lattice size, which is now run — is the obvious next requirement. D_{norm} ’s absolute scale is doubly qualified: it moves with N (§4) and its numerator’s coupling is not extremal (§5), so only its relative behaviour is quoted. Census ground truth is synthetic only. **Statistical scope.** The developmental family is pre-registered with BH-FDR at the run level and the ECA test uses the rule as its unit; λ_{ca} statistics exclude runs whose damage never ignited, for which the quantity is undefined, with n stated. The ladder reproductions are exact or near-exact matches to known values and rest on no hypothesis test. The cross-level negative rests on sign-consistency, not a surviving p -value. A previously reported capacity $\rightarrow$ sensitivity axis was pseudoreplicated and is retracted.

9 Responsible use and conclusion

This instrument measures a construction built *from* a language model, using only sampled outputs. Its intended uses are black-box: characterising models behind an API where weights are unavailable, and detecting developmental phase changes across checkpoints. It creates no new generative capability and needs no privileged access. The misuse we consider most likely is a misreading of our own numbers, which is why §5 is in the body rather than an appendix: a damping length or light cone measured here describes an in-place-resampling ring, and real generation absorbs *no* injected

token error. These quantities must **not** be cited as evidence about the robustness, error tolerance or self-correction of a deployed decoder — a λ_{ca} below zero is not a safety property. All models and corpora used are publicly released and used within their licences. That boundary is the point: an instrument that reproduces what is independently known, says which of its checks are load-bearing, and marks where its readings stop applying is more useful than one that claims more.

References

- Julian Arnold, Flemming Holtorf, Frank Schäfer, and Niels Lörch. Phase transitions in the output distribution of large language models, 2024.
- Franco Bagnoli, Raúl Rechtman, and Stefano Ruffo. Damage spreading and lyapunov exponents in cellular automata. *Physics Letters A*, 172(1–2):34–38, 1992.
- Nils Bertschinger and Thomas Natschläger. Real-time computation at the edge of chaos in recurrent neural networks. *Neural Computation*, 16(7):1413–1436, 2004.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, et al. Pythia: A suite for analyzing large language models across training and scaling, 2023.
- Eytan Domany and Wolfgang Kinzel. Equivalence of cellular automata to Ising models and directed percolation. *Physical Review Letters*, 53(4):311–314, 1984.
- Jesseba Fernando and Grigori Guitchounts. Dynamics of the transformer residual stream: Coupling spectral geometry to network topology, 2026.
- Jackson George, Zachariah Yusaf, Stephanie Zoltick, and Linh Huynh. (un)biased data and spin glasses reveal clustering for Turing phase transitions within human–transformer interactions, 2025.
- Peter Grassberger. Are damage spreading transitions generically in the universality class of directed percolation? *Journal of Statistical Physics*, 79(1):13–23, 1995.
- James E. Hanson and James P. Crutchfield. Computational mechanics of cellular automata: An example. *Physica D: Nonlinear Phenomena*, 103(1–4):169–189, 1997.
- Haye Hinrichsen, Joshua S. Weitz, and Eytan Domany. An algorithm-independent definition of damage spreading—application to directed percolation. *Journal of Statistical Physics*, 88(3): 617–636, 1997.
- G. A. Kohring and M. Schreckenberg. The Domany–Kinzel cellular automaton revisited. *Journal de Physique I*, 2(10):2033–2037, 1992.
- Christopher G. Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3):12–37, 1990.
- Xiaojuan Li et al. Cognitive activation and chaotic dynamics in large language models: A quasi-lyapunov analysis of reasoning mechanisms, 2025.
- Elliott H. Lieb and Derek W. Robinson. The finite group velocity of quantum spin systems. *Communications in Mathematical Physics*, 28(3):251–257, 1972.
- Shu-Wei Liu, J. Willsher, T. Bilitewski, Jin-Jie Li, A. Smith, K. Christensen, R. Moessner, and J. Knolle. Butterfly effect and spatial structure of information spreading in a chaotic cellular automaton. *Physical Review B*, 103(9):094109, 2021.
- Thomas McGrath, Matthew Rahtz, Janos Kramar, Vladimir Mikulik, and Shane Legg. The hydra effect: Emergent self-repair in language model computations, 2023.
- Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Critical phase transition in large language models, 2024.
- Prashanti Nilayam and Sankalp Nayak. Quiver: A formal framework for quantifying perturbation propagation and bifurcation in compound ai systems, 2026.

- David N. Olivieri and Antonio F. Pérez Rodríguez. Transformer field theory: A response-theoretic approach to mechanistic interpretability, 2026.
- Jérémy Perez, Grgur Kovač, Corentin Léger, Cédric Colas, Gaia Molinaro, Maxime Derex, Pierre-Yves Oudeyer, and Clément Moulin-Frier. When LLMs play the telephone game: Cultural attractors as conceptual tools to evaluate LLMs in multi-turn settings, 2024.
- Ingrid Petrova and Luan Vejsiu. Spectral origins of the self-correction blind spot in autoregressive generation, 2026.
- Huajian Ruan, Jinyang Li, Xingyu Guo, and Lingxiao Wang. Generative criticality in large language model temperature scaling, 2026.
- Cody Rushing and Neel Nanda. Explorations of self-repair in language models. In *International Conference on Machine Learning (ICML)*, 2024.
- Suvadip Sana, Sami Wolf, Neer Mehta, Alina Shah, Aitzaz Shaikh, Janna Goodman, and Lionel Levine. Mixing times of glauher dynamics on masked language models, 2026.
- Youran Sun and Babak Haghighat. Phase transitions in large language models and the $o(n)$ model, 2025.
- Luhan Tang, Longxuan Yu, Shaorong Zhang, and Greg Ver Steeg. Is your diffusion sampler actually correct? a sampler-centric evaluation of discrete diffusion language models, 2026.
- Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic perspective from jacobians, 2025.
- Keishu Utimula. Logos-ca: A cellular automaton using natural language as state and rule, 2026.
- Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing, 2025.
- G. F. Zebende and T. J. P. Penna. The Domany–Kinzel cellular automaton phase diagram. *Journal of Statistical Physics*, 74(5):1273–1279, 1994.
- Shiyang Zhang, Aakash Patel, Syed A. Rizvi, Nianchen Liu, Sizhuang He, Amin Karbasi, Emanuele Zappala, and David van Dijk. Intelligence at the edge of chaos. In *International Conference on Learning Representations (ICLR)*, 2025.

A Reproducibility

Models. bert-tiny/mini/base ($L=2-24$); Pythia-14M-1B [Biderman et al., 2023] and GPT-2 small-xl; Pythia-410m checkpoints step256–step143000. A tiny transformer trained from scratch on tinyshakespeare provides the corpus-checked census. **Grids.** $r \in \{1, 2, 4, 8, 16\}$, $T \in \{0.3, \dots, 1.5\}$, $N \in \{48, 96, 192, 384\}$, 8 seeds for the developmental family. **Definitions.** $D_{\text{norm}} = D/D_0$ with D_0 the independent-noise no-flip floor; λ_{ca} the log-slope of damaged-site count over the pre-saturation window, with the fit window’s constants exposed as parameters and a named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every back-end and both update modes through one shared simulation loop; a golden-file regression asserts the core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control that must fail. All code, per-run result files, and figure scripts accompany the submission; every number in the paper is traceable to a result file.

NeurIPS Paper Checklist

1. Claims

Question: Do the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract states the contribution as a *validated* black-box instrument and names the three things it does: reproduce known metrics before measuring LMs (the load-bearing rung being Domany–Kinzel, where agreement is *bit-exact* rather than fitted), detect a developmental transition across Pythia checkpoints, and *delimit* itself — real generation absorbs no injected token error, so the quantities describe an iterated-resampling construction rather than decoding. The smooth-limit rungs (logistic map, coupled-map lattice) are labelled in the abstract and body as arithmetic unit tests, not validation. The criticality phenomena are framed as decades-old (measured, not discovered) and cited to prior work. The cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed — that earlier two-seed gap is retracted and explicitly not built upon.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: The Limitations section states: neither construction samples the model’s actual distribution (masked conditionals are globally inconsistent; the autoregressive rule is consistent but resampled in place on a ring), so all claims are properties of the dynamics; small rings ($N \leq 384$) and one model family for the developmental result, with a second family named as the obvious next requirement; D_{norm} ’s absolute scale doubly qualified (it moves with N , and its numerator’s coupling is not extremal), so only its relative behaviour is quoted; synthetic-only census ground truth; the windowing scheme as a first-class apparatus factor; temperature scales not comparable across constructions. The statistical scope is stated separately, including that λ_{ca} statistics exclude runs whose damage never ignited — for which the quantity is undefined — with n stated in each case.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are supplied as anonymised supplementary material; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: The anonymised supplement contains the full codebase with a README giving per-phase reproduction commands and measured M1 runtime tables, pinned dependency versions, and idempotent resumable scripts. Every number in the paper is traceable to a committed result file; the analysis scripts assert their emitted group sizes against their declared pre-registered design. A public repository URL will be added at camera-ready if the paper is accepted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps}, \text{seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. Experiment statistical significance

Question: Does the paper report error bars / statistical significance?

Answer: [Yes]

Justification: The paper’s central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed ($p=0.470$), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p -value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. Broader impacts

Question: Does the paper discuss potential positive and negative societal impacts?

Answer: [Yes]

Justification: See the Responsible use section. Positive: black-box measurement of trained-LM token dynamics without weight access, applicable to closed/API-only models. It creates no new generative capability. The negative case we treat as most likely is misreading: the paper’s damping length and light cone describe an in-place-resampling construction, and real autoregressive generation absorbs no injected token error, so they must not be cited as evidence about deployed-decoder robustness. That boundary is stated in the body, not an appendix.

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk assets?

Answer: [NA]

Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. Licenses for existing assets

Question: Are existing assets (code, data, models) properly credited and licenses respected?

Answer: [Yes]

Justification: All assets are publicly released and cited: Pythia and BERT checkpoints (Apache-2.0), GPT-2 (MIT), tinyshakespeare (public-domain source text), WikiText-103 (CC BY-SA 3.0), PyTorch (BSD-3-Clause), HuggingFace Transformers and JAX (Apache-2.0). Usage is research-only and within each licence.

13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

404 Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy
405 checkpoints are documented in the repository README and the Reproducibility appendix.

406 **14. Crowdsourcing and research with human subjects**

407 Question: N/A.

408 Answer: [NA]

409 Justification: No human subjects or crowdsourcing.

410 **15. Institutional review board (IRB) approvals**

411 Answer: [NA]

412 Justification: No human-subjects research.

413 **16. Declaration of LLM usage**

414 Question: Does the paper describe LLM usage if it was an important, original, or non-
415 standard component?

416 Answer: [Yes]

417 Justification: Pretrained LLMs are the object of study — they supply the CA update rule —
418 and this is documented throughout. Separately, and disclosed here per venue policy: an AI
419 coding assistant was used substantially in implementing the experiments, in analysis script-
420 ing, and in drafting. All experimental designs, pre-registrations and claims were reviewed
421 and verified by the authors against committed result files; several errors introduced during
422 that process were caught by the repository’s own assertions and are recorded as retractions
423 in the supplementary findings log.